>> Intelligent Systems – WC 5<<
Vagueness, Probabilities and Bayesian Networks

Concepts to be known after WC5 :

- Reasoning with Vagueness
- Fuzzy Logic
- Probabilistic Knowledge
    - Independent/dependent events (+ calculation rules)
    - Probability
        - Joint
        - prior ('unconditional')
        - posterior ('conditional')
    - Bayes' theorem
- Bayesian Networks

Exercises Vagueness, Fuzzy Logic and Fuzzy Relations

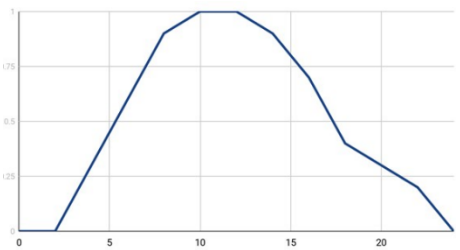
Exercise 1: Consider the following statement: "Half of the Dutch men are married". Which of the following statements is true?

A) The author of this statement describes the marital status of Dutch men with a degree of vagueness.
B) The author of this statement describes the marital status of Dutch men with a degree of uncertainty.

Give an example for a statement for the other property?
Dutch men are very tall.

Exercise 2: The director of our institute wants to reduce the environmental impact of the company by reducing the CO2 emissions caused by transportation. Therefore, she wants to give a devise a policy to support those who do not live very close (they can come by bicycle) or those who live very far (they should do remote work).

Since you know the director is a big fan of fuzzy set theory, you decide to model this problem with fuzzy sets. How do you model the fuzzy set of "employees who live moderately far away", i.e. those who do not live close, but also not far away?



Exercise 3: Given the following (standard) semantics for Fuzzy Logic for the following operators:

- f(¬A) = 1 - f(A)
- f(A v B) = max(f(A);f(B))
- f(A & B) = min(f(A);f(B))
- f(A -> B) = min(1;1 – f(A) + f(B))

What is the meaning of f(((A -> B) v B) & (¬A v ¬B)) if f(A)=0.8 and f(B)=0.5?

f(A -> B) = min(1 ; 1 - f(A) + f(B)) = min(1 ; 1 - 0.8 + 0.5)= min(1 ; 0.7) = 0.7

f((A -> B) v B) = max(f(A->B) ; 0.5) = max(0.7 ; 0.5)= 0.7

f(-A v -B) = max(1 - f(A); 1 - f(B)) = max(0.2 ; 0.5) = 0.5

f(((A -> B) v B) & (-A v -B)) = min(f((A -> B) v B) ; f(-A v -B)) = min(0.7 ; 0.5) = 0.5

Exercise 4: Suppose that there are two relations defined as follows, where L5 and L6 stand for Lecture 5 and 6 respectively:

understands

	L5	L6
John	0.4	0.6
Peter	0.3	0.7
Julia	0.1	0.4

contains

	semantics	reasoning
L5	0.8	0.3
L6	0.4	0.8

Fill in the table for the composition of the two relations, i.e. the relation

understandslecturecontaining = understands o contains

	semantics	reasoning
John		
Peter		
Julia		

You need to apply the following formula for each of the combinations:

$$f_{R\square S}(\langle x,z\rangle) = \max_{y\in Y} \min(f_R(\langle x,y\rangle), f_S(\langle y,z\rangle))$$

So,$f_{\text{understands o contains}}$ (<John,semantics>)

$= \max_{L5,L6} \min(f_{understands}(John,L5),f_{contains}(L5,semantics)),$
$\min(f_{understands}(John,L6),f_{contains}(L6,semantics))$
$= \max_{L5,L6} \min(0.4,0.8), \min(0.6,0.4) = \max_{L5,L6} 0.4,0.4 = 0.4$

And the same for the other fields:

John, reasoning = max(min(0.4,0.3),min(0.6,0.8))=0.6 -> this is informal
Peter, semantics = max(min(0.3,0.8),(0.7,0.4))= 0.4
Peter, reasoning = max(min(0.3,0.3),(0.7,0.8))=0.7
Julia, semantics = max(min(0.1,0.8),min(0.4,0.4))=0.4
Julia, reasoning = max(min(0.1,0.3),min(0.4,0.8)) = 0.4

	semantics	reasoning
John	0.4	0.6
Peter	0.4	0.7
Julia	0.4	0.4

Exercises Probabilistic Knowledge

Exercise 5: Consider the full joint probability distribution P(A,B,C) shown in the table to the right:

a) What is P(A = true)? 0.57 (sum of 0.15, 0.02, 0.07, 0.33)
b) What is P(B ^ C = true)? 0.58 (sum of 0.25 and 0.33)

A	B	C	P(A, B, C)
false	false	false	0.05
false	false	true	0.10
false	true	false	0.03
false	true	true	0.25
true	false	false	0.15
true	false	true	0.02
true	true	false	0.07
true	true	true	0.33

Exercise 6:

	toothache		*¬ toothache*	
	catch	*¬ catch*	*catch*	*¬ catch*
cavity	.108	.012	.072	.008
¬ cavity	.016	.064	.144	.576

Given the full joint distribution as above, calculate the following:

a. P(toothache).
b. **P**(Cavity).
c. **P**(Toothache|cavity).
d. **P**(Cavity|(toothache v catch)).

The main point of this exercise is to understand the various notations of bold versus non-bold P, and uppercase versus lowercase variable names. The rest is easy, involving a small matter of addition.

a. This asks for the probability that Toothache is true. P(toothache) = 0.108 + 0.012 + 0.016 + 0.064 = 0.2
b. This asks for the vector of probability values for the random variable Cavity. It has two values, which we list in the order ⟨true , false⟩. First add up 0.108 + 0.012 + 0.072 + 0.008 = 0.2. Then we have P(Cavity) = ⟨0.2, 0.8⟩.
c. This asks for the vector of probability values for Toothache, given that Cavity is true. Do not forget to normalize the values, the sum should be one.
   P(A|B) = P(A,B) / P(B)
   P(Toothache |cavity) = ⟨ (.108 + .012)/0.2, (0.072 + 0.008)/0.2⟩= ⟨0.6, 0.4⟩
d. This asks for the vector of probability values for Cavity, given that either Toothache or Catch is true. First compute P(toothache v catch) = 0.108+0.012+0.016+0.064+ 0.072 + 0.144 = 0.416.
   Then P(Cavity|toothache v catch) = ⟨(0.108 + 0.012 + 0.072)/0.416, (0.016 + 0.064 + 0.144)/0.416⟩= ⟨0.4615, 0.5384⟩

Exercise 7:

You throw two fair dice, one green and one red, and observe the numbers uppermost. Decide whether the following pairs of events are (in)dependent.
A: the sum is even
B: the red die is even

Independent.
Explanation.
P(A) = ½ (because 9 times even+even {(2,2),(2,4),(2,6),…} and 9 times odd+odd {(1,1),(1,3),(1,5),…} from all 36 possible outcomes)

P(A | B) = ½ (because 9 possible outcomes out of 18 possibilities in total). The occurrence of B does not change the probability of event A, so the events are independent. IN other words P(A|B)=P(A)
Or: two events A and B are independent if and only if their joint probability equals the product of their probabilities: P(A,B)=P(A) P(B)

Exercise 8:
You throw two fair dice, one green and one red, and observe the numbers uppermost. Decide whether the following pair of events are (in)dependent.
A: the sum is 5
B: the red die shows a 2
Dependent.
Explanation:
P(event 1) = 4/36 = 1/9 (1+4, 4+1, 2+3, 3+2)
P(event 1 | event 2) = 1/6 (now only one throw of the green dice makes event 1 true, namely a 3).
The occurrence of event B changes the probability of event A, so the events are dependent.

Exercise 9:
Given the following information:

- The probability that someone with a cold also has fever is 80%.
- The prior probability of having a cold is 0.0001.
- More or less 1 out of 1000 people in the population has fever.

I have fever. Calculate with Bayes' Rule the probability that I have cold.

Answer:
P(fever|cold) = 0.8
P(cold) = 0.0001
P(fever) = 0.001
P(cold|fever) = (P(fever|cold) x P(cold)) / P(fever) = (0.8 x 0.0001) / 0.001 = 0.08
The probability that I have a cold given I have fever is 8%

Exercise Bayesian Network

Exercise 10

B	P(B)
+b	0.001
-b	0.999

E	P(E)
+e	0.002
-e	0.998

B	E	A	P(A|B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

A	J	P(J|A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

A	M	P(M|A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

Given the following Bayesian network, what is the

- Probability that an earthquake happens?
  - A: Earthquake is independent of other probabilities, so it comes directly from the table, +e = 0.002
- probability that the alarm rings?
  - A: We are looking for the probability that the alarm rings, i.e., +a
  - The alarm is dependent on whether a burglary and/or a burglary is happening. It does not depend on whether Joh calls, or Mary calls.
  - The probability that the alarm rings, when there is, or there is no, earthquake and there is, or there is no, burglary is:
    - P(a+, b+/-,e+/-) = P(a+, b+, e+) + P(a+, b+, e-) + P(a+, b-, e+) + P(a+, b-, e-)
  - So, we need to sum the probabilities for all situations in which the alarm rings.
  - Let's start with the first one, the others are analog
    - P(a+|b+,e+) = 0.95 , that means that the probability that the alarm rings, given that there is both a burglary and an earthquake is 0.95.
    - Now, this assumes that an earthquake is happening and also a burglary is happening. These events have a probability of 0.002 and 0.001, respectively. So, the probability that both happen at the same time, is
      - 0.002*0.001 = P(b+, e+)
    - So, the probability that the alarm rings because of both a burglary happening and an earthquake happening is
      - P(b+, e+) * P(a+|b+,e+) = 0.002*0.001*0.95 = 0,0000019

            - An alternative is to apply the formula from the lecture: P(a+, b+, e+) = P(a+, b+, e+) = P(a+| b+, e+) * P(b+| e+) * P(e+) = P(+a|b+,e+)*P(b+)*P(e+)= 0.002*0.001*0.95 = 0,0000019
        -
    - Analog, we find
        - P(b+, e-) * P(a+ | b+, e-) = 0.001 * 0.998 * 0.94 = 0.00093812
        - P(b-, e+) * P(a+ | b-, e+) = 0.999 * 0.002 * 0.29 = 0.00057942
        - P(b-, e-) * P(a+ | b-, e-) = 0.999 * 0.998 * 0.001 = 0.000997002
    - Summing these probabilities together, we find that the probability that the alarm rings is 0,002516442
- Probability that john calls?
    - A: this is directly derivable from the previous result since the probability that John calls is only dependent on the alarm.
    - The probability that John calls is the probability that he calls while there is an alarm plus the probability that he calls while there is no alarm.
        - P(j+, a+/-) = P(j+,a+) + P(j+,a-)
    - P(j+,a+) = P(a+)* p(j+|a+) = 0,002516442 * 0.9 = 0,0022647978
    - P(j+,a-) = P(a-)*p(j+|a-) = (1-0,002516442) * 0.05 = 0,0498741779
    - So, the total probability that John calls is 0,0521389759